

BOBA_DATABASE

Boba System Database

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The boba-jackett Go service (port 7189) owns a SQLite system database at config/boba.db (default; override with BOBA_DB_PATH). Sensitive columns are encrypted with BOBA_MASTER_KEY (AES-256-GCM).

This document is the canonical reference for the schema, the master key lifecycle, backup, file permissions, and recovery from corruption.

1. Schema reference

Defined in qBitTorrent-go/internal/db/migrations/001_init.sql. See the spec at docs/superpowers/specs/2026-04-27-jackett-management-ui-and-system-db-design.md § 5 for full normative text.

Table	Purpose	Encrypted columns
credentials	Per-tracker credential bundle. kind is userpass or cookie. Holds username_enc, password_enc, cookies_enc (any combination per kind). Tracks created_at, updated_at, last_used_at.	username_enc, password_enc, cookies_enc
indexers	Local registry of indexers configured at Jackett. Each row points (optionally) at a	none

Table	Purpose	Encrypted columns
	credentials row by linked_credential_name. Tracks enabled_for_search, last_jackett_sync_at, last_test_status, last_test_at.	
catalog_cache	Snapshot of Jackett's indexer catalog (last refresh). UI browse + filter source. template_fields_json is the per-indexer config template fetched from Jackett.	none
autoconfig_runs	Append-only history of every auto-config run. Stores ran_at, counts, redacted result_summary_json, errors_json. Indexed by ran_at DESC for the UI runs page.	none (summary is already redacted)
indexer_map_overrides	Explicit env_name → indexer_id map. Highest precedence in the matcher (beats fuzzy matching).	none
schema_migrations	Schema version tracking; row 1 corresponds to 001_init.sql.	none

Indexes

- idx_indexers_linked_cred ON indexers(linked_credential_name)
- idx_catalog_cached_at ON catalog_cache(cached_at)
- idx_runs_ran_at ON autoconfig_runs(ran_at DESC)

Pragmas

The DB is opened with WAL journaling, foreign keys on, and busy-timeout sufficient for the bootstrap import + UI write rate.

2. Master key lifecycle

Auto-generation (canonical path)

BOBA_MASTER_KEY is a 32-byte hex AES-256-GCM key. It is generated **exactly once**, on first boot, by `internal/bootstrap.EnsureMasterKey` (see `qBittorrent-go/internal/bootstrap/bootstrap.go`).

`start.sh` also calls an `ensure_boba_master_key` helper as a belt-and-suspenders so the variable is present in the host's `.env` even if the container has not yet booted.

The two-step write (key generated in memory, then written to `.env`) was collapsed into a single atomic `envfile.AtomicWriteAtCommit` `c0f20bc` to eliminate a silent data-loss window. The regression guard is `TestEnsureMasterKeyDoesNotDuplicateHeader` in `qBittorrent-go/internal/bootstrap/bootstrap_test.go`.

Key Rotation

STATUS (verified 2026-08-20): NOT EXECUTABLE AS WRITTEN.

Both subcommands below — `rotate-key` and `envfile-replace` — are **absent from the source**. Verified by a control-needle-checked search of `qBittorrent-go/**/*.go` (§11.4.201(7)(b): the needle `EnsureMasterKey` matched, so the absences are real, not a blind `grep`). There is also no built `bin/boba-jackett` to probe. Treat the block below as the INTENDED design, not a working runbook: today, use the “Loss recovery” path with `.env` as the source of truth. Tracked as `TODO(BOB_MASTER_KEY_ROTATION)`.

Heading note: this section is named “**Key Rotation**” because `internal/bootstrap/bootstrap.go` writes `# To rotate: see docs/BOBA_DATABASE.md § "Key Rotation"` into every operator's `.env`. It was previously titled “Manual rotation”, so that shipped pointer resolved to nothing.

To rotate the master key without losing credentials (INTENDED design):

```

# 1. Stop the service (use the orchestrator, never raw podman/
    docker)
make boba-jackett-stop # or your orchestrator's equivalent

# 2. Decrypt all rows with the OLD key, re-encrypt with a NEW key.
#    Use the rotate subcommand:
./bin/boba-jackett rotate-key \
    --db /config/boba.db \
    --old-key "$OLD_KEY" \
    --new-key "$NEW_KEY"

# 3. Atomically update .env (envfile.Atomic uses tmpfile + fsync +
    rename)
./bin/boba-jackett envfile-replace BOBA_MASTER_KEY "$NEW_KEY"

# 4. Start the service
make boba-jackett-start

```

(Implementation note: a rotate-key subcommand is the canonical place for this operation; if not yet present in the binary, follow the “loss recovery” path below using .env as the source of truth.)

Loss recovery

There is **no recovery** from BOBA_MASTER_KEY loss alone. Encrypted rows in credentials become permanently undecipherable.

The only recovery path: the original <NAME>_USERNAME / <NAME>_PASSWORD / <NAME>_COOKIES triples in .env (or wherever they were originally entered) are still present. To re-bootstrap:

```

# 1. Stop the service
# 2. Move the now-useless DB out of the way
mv config/boba.db config/boba.db.lost
# 3. Generate a new master key (or let bootstrap do it)
sed -i '/^BOBA_MASTER_KEY=/d' .env
# 4. Start the service – bootstrap.EnsureMasterKey will generate a
    fresh
#    key and re-import credentials from .env into a brand new
    boba.db.
make boba-jackett-start

```

UI-only edits made after the original bootstrap are NOT recoverable — they were never written back to .env.

3. Backup procedure

Performable as of 2026-08-21. This procedure was previously impossible to carry out, and the document did not say so — it prescribed no workaround because it never recorded the blockage. `config/boba.db` is mode `600`, and its owner was a rootless-container sub-uid (`100999`) with no host account, so the operator could not read their own credential database and neither `cp` nor `sqlite3 .backup` could open it. Feature `002-user-owned-downloads` restored the file to the operator. **Measured 2026-08-21:**

```
config/boba.db mode=600 owner=milosvasic(1000)
group=milosvasic(1000) size=81920
```

A copy of the file read cleanly (81920 bytes) and the source stayed mode `600` afterwards. No elevation, no `podman unshare`, and no `chown`-first step is needed — if you find yourself reaching for one, the ownership regression has returned: run `scripts/ownership_precondition.sh` (it names the offending location) and then `scripts/ownership_repair.sh`. See [scripts/ownership_repair.md](#) and [guides/file-ownership.md](#).

Two acceptable approaches.

Cold backup (service stopped):

```
make boba-jackett-stop
cp config/boba.db config/boba.db.bak
cp .env .env.bak
make boba-jackett-start
```

Hot backup (service running): use SQLite's online `.backup`:

```
sqlite3 config/boba.db ".backup config/boba.db.bak"
cp .env .env.bak
```

`.backup` respects WAL semantics and produces a transactionally consistent snapshot. **Always back up `boba.db.bak` and `.env.bak` together** — the DB is useless without the matching master key, and the master key is useless without the DB.

Both files **MUST** be stored with mode `0600` (or stricter) on persistent media; treat them as you would a private SSH key.

The fix restored the OWNER, not the mode — and the mode must stay 600

config/boba.db remains mode 600 and **MUST** stay mode 600. The ownership repair changes who owns the file; it explicitly does not relax who may read it (the file is declared `preserve_mode: true` in config/owned_paths.yaml, and scripts/ownership_repair.sh restores each item's original permission bits after the `chown`, because a non-root `chown(2)` can clear them).

This distinction is the whole point. A backup that started succeeding because the file had become world-readable would not be a fix — it would be a security regression wearing a fix's clothes: a credential store made readable by every account on the host, traded for a convenience the correct owner already provides. The backup works now because the file **belongs** to the operator, not because it was opened up.

If `stat -c '%a %U' config/boba.db` ever reports anything wider than 600, or an owner that is not the account running the system, stop and treat it as a defect. The standing guards are the Layer 4 challenge `boba_db_file_perms_challenge.sh` (§ 4 below) for the mode, and pre-build invariant 33 CM-OWNERSHIP-INVARIANTS for the ownership route.

4. File permissions

Both files **MUST** be 0600 (owner read/write only):

File	Enforced by	Code reference
config/boba.db (and -wal, -shm siblings)	db.Open chmods on first open if mode is wider	qBitTorrent-go/ internal/db/conn.go line ~53
.env	internal/ envfile.Atomic writes with mode 0600	qBitTorrent-go/ internal/envfile/ write.go

The Layer 4 security challenge `boba_db_file_perms_challenge.sh` is the regression guard — it **MUST PASS** before any release. The corresponding Go test is `qBitTorrent-go/internal/db/credential_leak_test.go`.

The DB file mode bug (modernc.org/sqlite creating boba.db 0644) was caught at commit 8405167 and fixed in db.Open with os.Chmod(path, 0o600). See also docs/issues/fixed/BUGFIXES.md.

5. Recovery from a corrupted DB

If boba.db becomes corrupted (truncated, partial writes, foreign-key violations from manual edits, etc.), the recovery path is:

```
# 1. Stop the service (orchestrator)
make boba-jackett-stop

# 2. Move the broken DB aside (DO NOT delete – keep it for
    forensics)
mv config/boba.db config/boba.db.broken
# Also move WAL/SHM if present
mv config/boba.db-wal config/boba.db-wal.broken 2>/dev/null ||
    true
mv config/boba.db-shm config/boba.db-shm.broken 2>/dev/null ||
    true

# 3. Restart – bootstrap.EnsureMasterKey notices the missing DB,
#     re-creates the schema (migration 001), then re-imports any
#     credentials still present in .env.
make boba-jackett-start

# 4. Verify
curl -s http://localhost:7189/healthz | jq .
curl -s http://localhost:7189/api/v1/credentials | jq 'length'
```

Anything entered via the UI after the original bootstrap and before the corruption is unrecoverable from .env alone — see § 2 “Loss recovery”. Always pair regular backups (§ 3) with this recovery path.

6. Related files

- Schema source: qBitTorrent-go/internal/db/migrations/001_init.sql
- Open / chmod logic: qBitTorrent-go/internal/db/conn.go
- Master key bootstrap: qBitTorrent-go/internal/bootstrap/bootstrap.go
- Atomic .env writer: qBitTorrent-go/internal/envfile/write.go
- Credential repo: qBitTorrent-go/internal/db/credentials.go
- Bug history: docs/issues/fixed/BUGFIXES.md

- Integration overview: docs/JACKETT_INTEGRATION.md § Auto-Configuration